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# **A Study of the Demutualization of the Philippine Stock Exchange and its Effect on Stock Return Volatility**

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## **INTRODUCTION**

Stock exchanges all over the world are introducing new ways to increase their capitalization. One of this is through demutualization, where the stock exchange is converted to stockholder based organization for profit taking. The earliest known demutualized stock exchange is in Stockholm in 1993. Through demutualization, the exchange can raise additional capital through the initial public offering. Also, with a demutualized stock exchange, it can compete globally by offering more products and accessing other markets.

The Philippine Stock Exchange, Inc., (PSE)—seeing that the trend is for the demutualization of stock exchanges—followed the Australian Stock Exchange model. The PSE was demutualized on August 8, 2001 as provided by Sec. 33 (a) of the Securities Regulation Code (SRC) of 2000. Based on a demutualized exchange, it is aimed to have better corporate governance, more

products traded, more investors participating in the market, and more companies enlisting in the exchange.

This study will determine how the demutualization of the PSE affects the stock return volatility of the Philippine Stock Exchange Composite Index (PHISIX) thirty days before and after demutualization.

The study will include the daily results of the PHISIX thirty days before and after the demutualization date of the PSE covering the period June 27, 2001-September 21, 2001. Only nominal daily Phisix will be used for the analyses. It was assumed that the daily index value truly reflects the market sentiment.

One limitation of the study is that stock returns were not adjusted for stock market anomalies such as weekend effect, holiday effect, and January effect.

### **PHILIPPINE STOCK EXCHANGE, INC.**

To remain globally competitive, the Philippine Stock Exchange, Inc. (PSE), has to find ways of improving the stock market and have more investor participation. The first step they undertook is to become demutualized. The conversion was made possible through a technical assistance grant from the Australian Agency for International Development, given through the Philippine Australia Governance Facility. As such, the ASX model was used as framework for the demutualization of the PSE.

### **AUSTRALIAN STOCK EXCHANGE (ASX) (FOR EQUITIES, DERIVATIVES, FIXED INTEREST SECURITIES)**

In 1987, six separate state based stock exchanges, namely, Sydney, Melbourne, Brisbane, Perth, Adelaide, and Hobart operating throughout Australia were unified under the Australian Stock Exchange (ASX). This created a national market and one body to govern stock market trading throughout the country. The ASX was originally a mutual organization of stockbrokers, like its predecessor state stock exchanges. The exchange was a company limited by guarantee with no shares, and was prohibited from paying profits; surplus had to be applied towards promoting objects of the exchange.

In the same year, Stock Exchange Automated Trading Systems (SEATS) was introduced. In 1990, all other trading floors were abolished and were conducted at SEATS. In 1994, the first

stage of an electronic clearing and settlement system (CHES) was introduced.

In 1998, the Australian Parliament legislated the demutualization of the ASX, the world's first exchange to be demutualized. Under the demutualized exchange, the regulatory and public interest responsibilities of exchanges as self-regulatory organization (SROs) as well as the exchange's accountability to the regulators, the Australian Securities and Investment Commission (ASIC) were expanded. There is a separation from stockbrokers' right to trade on an exchange from shareholder right. With the demutualization, the ASX is committed to serve its shareholders which included its stockbrokers, customers, government, and the community. Thus, its expanding membership reached 606 with issuance of shares amounting to 166,000, priced at A\$4/share. However, a 5% limit on shareholdings in the exchange was imposed. Also, the exchange is allowed to self-list and provided for supervision of this arrangement by ASIC. The ASIC had to accept a way that would act as a supervisor for the companies that list on the exchange. This meant that the ASX could not effectively supervise itself. Although the ASX is a self regulatory organization, it needed the ASIC to act as a regulator for the ASX, making sure the exchange rules and policies are reinforced and followed by the members, brokers, and listed companies. As part of its corporate governance, the ASX will have nine-member board of directors.

The full change in status only took place on 13 October 1998. Listing of ASX and quotation of share occurred on 14 October 1998. The ASX converted from company limited by guarantee to public corporation.

In 1999, the ASX formed an alliance with the Nasdaq Stock Market. These two exchanges will work on co-listing or co-trading. Co-trading acts as a portal for investors in its environment to tap into securities trading in another market and vice versa. This will allow Australians to invest through ASX in Nasdaq's benchmark technology sector through a single vehicle. It will allow Australian investors, operating through an Australian broker and the ASX, to route orders into the US market and have that transaction executed there. The co-listing covers two situations. In co-listing, a firm can raise capital on one market only, but may list and trade in both markets from the time of the capital raising. Or the firm can raise capital on both markets. With two markets, transaction costs become less expensive and easier.

Aside from the Nasdaq, the ASX has established formal relationships with other Asian stock exchanges: Singapore, Malaysia, Korea, Indonesia, Philippine, Thailand, Taiwan. This will enable Australian investors to buy shares in the international markets using the ASX services.

*Note: Above discussion was taken from the "APEC FINANCIAL REGULATORS TRAINING INITIATIVE REGIONAL SEMINAR – DEMUTUALIZATION OF EXCHANGES" 13-14 August 2001, Asian Development Bank, Manila, Philippines. The talk on "The conversion to a demutualized exchange—ASX's experience" was made by David Holthouse, National Manager, International Affairs, Australian Stock Exchange.*

The demutualization of the PSE was in compliance with the 2000 Securities Regulation Code (SRC), Sec. 33. On August 8, 2001, the Philippines Stock Exchange (PSE) formally launched its conversion from a mutual company to a stock corporation. With this development, the PSE is no longer the so-called "Old Boys Club" used to be dominated mostly by member-brokers. This so-called fraternity has led to some alleged anomalies in the stock market trading, particularly the BW case in year 2000.

According to the provisions of the Securities Regulation Code (SRC) of the Philippine Stock Exchange, the Exchange restructured its Board to include seven (7) brokers and eight (8) non-brokers, one of whom is the President. The principle of sound corporate governance calls for a Board that acts as an effective oversight committee that safeguards the interests of all shareholders. In addition, it also manages risk, and to keep the legislation current with evolving standards.

The conversion to a stock corporation shall be effected by allocating a portion of the members' contribution in the amount of Php9,200,000.00 to the members of the Exchange. Based on the financial statements of the Exchange as of October 31, 2000, the members have a total contribution of Php286,627,000.00. The balance of the Members's Contribution account of Php277,427,000.00 shall be treated as additional paid-in surplus in the financial statements and will be distributed proportionally in the future.

The authorized capital stock of the corporation is Php36,800,000.00 divided into 36,800,000 shares at a par value of Php1.00 per share. The subscribed capital is Php9,200,000.00

**which shall be fully paid. Each member broker will have 50,000 shares at a par value of PhP1.00 per share of the capital stock.**

**Upon demutualization, the 184 member brokers was conferred one trading right in recognition of their ownership seats in the exchange. This trading right is transferable without any time limitation. Unlike the ASX, the exchange opted not to buy back the seats prior to demutualization.**

**The trading rights will allow the holders to trade at the exchange. Upon becoming a trading participant, a "Certificate of Trading Rights" shall be issued to its former members. The trading right shall be transferrable without any time limitation. However, trading rights will be limited to the existing one hundred eighty-four (184) members. Given prevailing market conditions and the thin trading volumes being shared by current players, the PSE has imposed an indefinite moratorium on the issuance of new trading rights. The PSE's conversion to a stock company represents only the initial phase of the demutualization process.**

**Currently fully owned by stock brokers, the PSE must diversify its ownership base according to an SRC provision which limits ownership of any one industry group to 20%. This reform measure was especially intended to address allegations that the PSE operated as an "old boys' club" incapable of policing its own ranks (a point highlighted by the BW Resources stock market scandal). The current challenge now lies in strengthening structures and profit centers in preparation for the PSE's envisioned public listing. SEC and PSE agreed on a time table for the bourse's listing but indicated that the parties were discussing a two-year time frame.**

## **RISK MANAGEMENT**

**In the Philippine capital market, the Securities Clearing Corporation of the Philippines (SCCP) handles the risk management of the Exchange. Currently, the PSE owns fifty-one (51%) of the SCCP. SCCP has acquired essential steps to enrich risk management capacities. Among the steps initiated are the reduction of settlement date from T+4 to T+3, continuous evaluation of the Comprehensive Trade Guaranty Fund (CTGF) to attain the ideal level, imposition of the Mark to Market Collateral Deposit (MMCD), and the creation of guidelines both for the cash payment or early delivery / early payment and special isolated immediate settlement (SIIS).**

In addition, daily monitoring of trading activity (**volume/value**) per Broker and per issue is in place. Actively traded issues are monitored based on the following:

1. Price band 40% floor; 50% ceiling
2. 50% increase in volume
3. Trade Value exceeds the 25% current level of CTGF
4. Equity Trades (EQTRADES)

## **BUSINESS OF THE EXCHANGE**

As of 9 July 2001, there are two hundred twenty nine (229) listed companies with a total market capitalization of approximately US\$46.7B. It continues to expand its product range and enhances the market infrastructure to expand its services as a center for capital formation. The following are the enhancements being made in the operations of the Exchange:

### ***1. Enhancement of Surveillance System***

The alerts to be introduced to the system and scheduled for implementation are:

1. **Cumulative Price Alert** - will list all issues deviating beyond a variance limit of cumulative price;
2. **Cumulative Volume Alert** - will list all issues deviating beyond a variance limit of cumulative volume;
3. **Cross Transaction Alert** - will list all brokers significantly involved in heavy cross transactions, compared to their regular floor transactions;
4. **Run-off Alert** -will list all significant trading activity conducted proximately prior to the run-off period (end of trading day);
5. **Significant Shares Concentration Alert** - will list all brokers, issuers who have had unusual concentration of traded shares activity; and
6. **Significant Trading Alert** - will list all issues with unusually significant market activity based on the issue's statistical average trading volume.

### ***2. Introduction of the Electronic Disclosure System***

The electronic disclosure system will allow listed companies to send their respective company disclosures (unstructured and structured) electronically. The company disclosures will be reviewed by the Listings and Disclosure Group, and broadcasted

to all market participants, data vendors and media via electronic data feed. The PSE website will also be made.

As of December 31, 2001, the revenues of the exchange were generated from the following:

1. Listing fees and other listing related activities (52%)
2. Information Technology Services (27%)
3. Interest Income (18%)
4. Membership (2%)
5. Administration Recoveries (1%)

The exchange is currently looking for opportunities to improve returns and minimize costs in the short term and studying the long-term strategic directions it will take.

*Note: Above discussion was taken from the "APEC FINANCIAL REGULATORS TRAINING INITIATIVE REGIONAL SEMINAR – DEMUTUALISATION OF EXCHANGES 13-14 August 2001, Asian Development Bank, Manila, Philippines. The talk on Demutualization of the Philippine Stock Exchange was made by Marie Larrie Alinsunurin.; (www.pse.org.ph; Philippine Stock Exchange Library)*

## **REASONS TO DEMUTUALIZED**

Based on the demutualization experiences of several stock exchanges (Australia, Singapore, Hong Kong, and Stockholm), there are common reasons for demutualizing. As defined earlier, demutualization is the conversion into a profitable stockholder-based entity. Aside from becoming a profit organization, there are several reasons for demutualizing: (1) raise capital to compete in new markets; (2) develop new state of the art technological platforms, and remain innovative through the offering of low-cost order routing and matching ; (3) to have a rationalized governance; (4) to attract investor participation; (5) to use share capital to fund new acquisitions; (6) Alliances and consolidations between exchanges are becoming wave of the future – scale and greater product variety are key to survival; (7) U.S. competitive threat Alternative Trading Systems (ATs); (8) the exchange to be considered as a business in its own right and at the same time, fulfill its responsibilities to its shareholders.

With demutualization, access to capital markets is through the initial public offering of additional securities. Each member of the exchange is allotted the same number of shares at a prescribed offering price. However, what is noticeable of the demutualized

stock exchanges is that no person or combination of persons acting jointly or in concert may beneficially own more than 5%. Those who will hold more than 5% need the consent of regulating bodies.

Today, almost all the stock exchanges is threatened by alternative trading systems (ATS), and increasing competition on both the domestic and international fronts. The exchange should be more responsive to the rapid demands for change which will better serve the interests of institutional, retail or individual brokers. One way of doing this is developing state-of-the art technology. This technology must be a robust, adaptable, and high-capacity technology platform to remain in a competitive position. This new technology will end trading capacity constraints that some exchanges have experienced in the past. These improvements confirm the exchanges' strong commitment to customers and shareholders to ensure a sound state-of-the-art and reliable trading system. With improved technology which information and transactions flow at speed and volumes that were beyond comprehension a decade ago. The most striking example of this has been in the rise of screen-based exchanges, which, in some sectors, are rapidly replacing floor-based exchanges. Thus, the exchanges must demutualize themselves in order for them to be globally competitive with other technology-driven exchanges. With such development, there is more efficiency in the market thus eliminating possible abnormal returns.

Also, with a demutualized exchange, it can form alliances with other markets thus giving investors more products to trade and ease of entry. Also, with more corporate governance and electronic disclosures, investors will have a better assessment of the true picture of the companies listed as well as the whole stock exchange. The "old boys club" thinking will eventually be erased. The "old boys club" refer to stock exchanges that are run like a membership organization. It is noted that these non-profit exchanges are too slow and too bureaucratic and it is too hard for the investors to make a decision. However, with the new set-up, investors will be able to participate in the market faster armed with efficient information and trust that there is transparency and better governance in the market.

## DISCUSSION

In 1950, Harry Markowitz's Portfolio Theory showed how to build portfolios of stocks with the highest possible expected return

given their risk or the lowest possible risk given their rate of return. He discovered that the expected return to a portfolio of stocks is an average of the returns to the stocks themselves. (Haugen, 1999).

Markowitz tried to figure out how to calculate the risk of a stock portfolio by using the standard deviation. The risk was defined, as the chance of getting returns different from one's expectation. The standard deviation of expected returns is a statistical measure of the dispersion of returns around the expected value whereby a larger variance or standard deviation indicates greater dispersion, all other factors being equal. The idea is the more dispersed the expected returns, the greater the uncertainty of those returns in any future period (Haugen, 1999). From the risk and return of each investment, a portfolio can be made to satisfy the needs of the investors.

In this research, the event study will be used. Event studies was first used by Dolley (1933) when he examined the price effects of stock splits, studying nominal prices changes at the time of the split. From 1921 to 1931, he found out that the price increased in 57 of the 95 cases and only 26 had prices that declined. There were other event studies that came forth (Myers and Bakay (1948), Ball and Brown (1968) and Fama, Fisher, Jensen, and Roll (1969).

Campbell, Lo, and MacKinlay (1998) gave an outline of the structure of an event study. The analysis can be viewed in several steps which this study will follow. The framework is given below.

**1. Event definition.** The initial task of conducting an event study is to define the event of interest and identify the period over which the security prices of the firms involved in this event will be examined — the event window. In this research the event is the demutualization of the Philippine Stock Exchange.

**2. Selection criteria.** After identifying the event of interest, it is necessary to determine the selection criteria for the inclusion of a given firm in the study. In this research, the Phisix will be used since it represents 30 stocks and is considered a representative of the performance of the stock market.

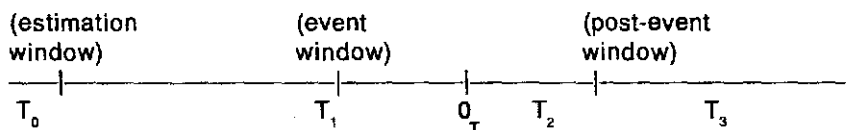
**3. Normal and abnormal returns.** To appraise the event's impact, a measure of the abnormal return is required. The abnormal return is the actual ex post return of the security

over the event window minus the normal return of the firm over the event window. The normal return is defined as the return that would be expected if the event did not take place. For each firm  $i$  and event date  $t$  we have

$$e_{it}^* = R_{it} - E(R_{it} | X_t), \quad (1.1)$$

where  $e_{it}^*$ ,  $R_{it}$ , and  $E(R_{it})$  are the abnormal, actual, and normal returns, respectively, for time period  $t$ .  $X_t$  is the conditioning information for the normal performance model. The constant-mean-return model where  $X_t$  is constant will be used.

- 4. Estimation procedure.** Once a normal performance model has been selected, the parameters of the model must be estimated using a subset of the data known as the estimation window. Generally, the event period itself is not included in the estimation period to prevent the event from influencing the normal performance model parameter estimates. In this case, we index returns in event time using  $t$ . Defining  $t=0$  as the event date,  $t=T_1+1$  to  $t=T_2$  represents the event window, and  $t=T_0+1$  to  $t=T_1$  constitutes the estimation window. Let  $L_1=T_1-T_0$  and  $L_2=T_2-T_1$  be the length of the estimation window and the event window respectively. The time line is give below:



- 5. Testing procedure.** With the parameter estimates for the normal performance model, the abnormal returns can be calculated. The abnormal return over the event window is a measure of the impact of the demutualization event on the value of the Phisix. In this study the period for estimation period is from June 27, 2001-August 7, 2001; event window is from August 8-9, 2001, and post-event window is from August 10- September 21, 2001.

Markowitz Portfolio Theory showed how to compute for the risk and return of a portfolio. In this research, the portfolio

will be represented by the Phisix for a 61 day period from June 27, 2001-September 21, 2001. Since the Phisix has thirty (30) stocks included in its basket, it is deemed as a diversified portfolio. The returns  $r_t$  of an asset is defined to be the natural logarithm of its gross return  $(1 + R_t)$ : (Campbell, Lo, Mackinlay, 1997, p. 11)

$$r_t \equiv \log (1 + R_t) = \log \frac{P_t}{P_{t-1}} \quad (1.2)$$

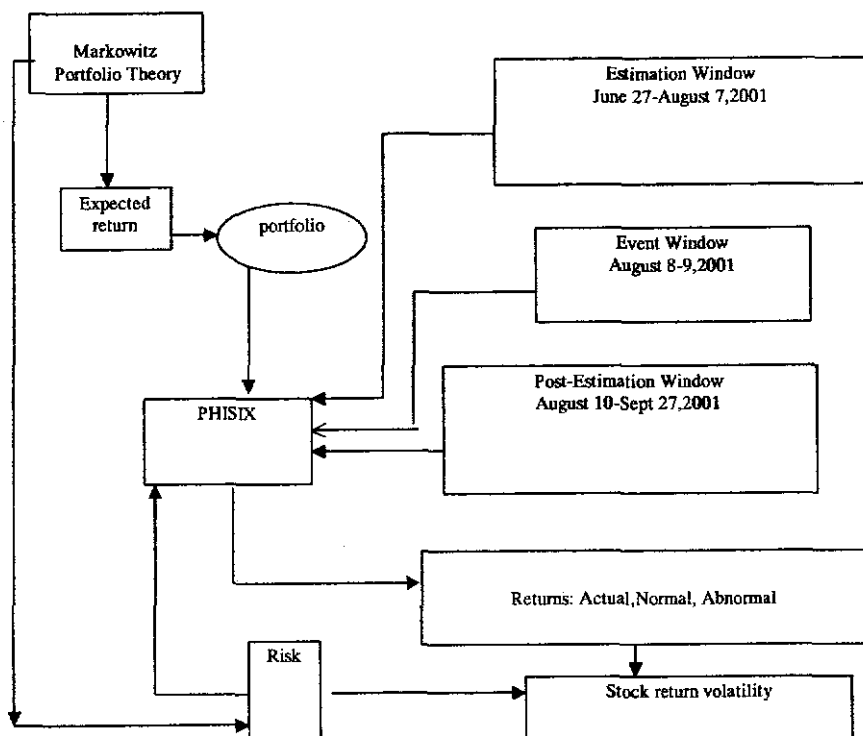
Returns to be computed included abnormal, actual, and normal returns. From the stock returns, its volatility can be measured through the standard deviation (see methodology section for computation).

**6. Empirical results.** The presentation of the empirical results follows the formulation of the econometric design.

**7. Interpretation and conclusions.** Ideally, the empirical results will lead to insights about the mechanisms by which the event affects security prices. From the computation of the abnormal, actual, and normal returns, further empirical tests will be made to be interpreted and analyzed.

#### **OPERATIONAL DEFINITION:**

1. Trading period is from 9:30 am to 12 noon with a ten-minute extension from closing time to execute orders at closing prices. During the ten minute run-off period, only volume may change but the closing price at 12 noon will remain unchanged.
2. **Philippine Stock Exchange Composite Index (PHISIX)**, aggregative measure of relative changes in the market capitalization of common stocks that provides a comprehensive picture of market trends. It is composed of a basket of thirty (30) common stocks of listed companies, selected to represent the general movement of the market prices. (PSE Fact Book, 1997). Selection of the companies to be included in the Phisix includes the following criteria: (1) liquidity; (2) market capitalization; (3) representation; (4) maturity and (5) stock type. Liquidity refers to the stocks that have been traded regularly with an



**Figure 1: Framework**

average daily value turnover of not less than five million pesos, excluding block transactions. Only those with the largest market capitalization stocks are included and they are representative of the industry. The stocks should also be traded for at least six months and only common stocks are eligible to be components of the indices. (**PSE FACT Book, 1999-2000**).

## DISCUSSION OF RESULTS

**Volatility before and after demutualization.** On the hypothesis of return variance before demutualization is equal to the return variance after demutualization, the f-test was used to validate this. There were two tests done. One is for thirty days before and thirty days after the demutualization date. The demutualization date is included in the sample size. The other test has a two-day lag after the demutualization date. From table 1 below, all the F-test resulted in values greater than 1. This means that there is a

significant difference in the volatility of stock return before and after demutualization. This is further evidenced by the standard deviation, which measures volatility of stock returns. Given the standard deviations before and after demutualization, Table 2 show that there is more volatility after demutualization.

**Table 1. F-test Results**

|                                                                  | Using Inreturns | Using abnormal returns |
|------------------------------------------------------------------|-----------------|------------------------|
| 30 days before/after demutualization including event date        | 3.13416E-05     | 5.57849E-05            |
| 30 days before/after demutualization, 2-day lag after event date | 6.46908E-05     | 6.46908E-05            |

**Table 2. Standard Deviation**

|                                                                  | Using Inreturns | Using abnormal returns |
|------------------------------------------------------------------|-----------------|------------------------|
| 30 days before/after demutualization including event date        | 0.006827        | 0.006827               |
| 30 days before/after demutualization, 2-day lag after event date | 0.015038        | 0.015236               |

However, in the sample period, 11 September 2001 was included. Another significant event which was the World Trade Center disaster caused by terrorist acts. This event significantly affected world financial markets which plummeted in the aftermath of the attack. Thus, there maybe a certain bias in the results. But a closer look at the returns for the period August 8-10 shows that they have improved from negative to positive, which means a favorable response from investors since they believe that through demutualization, there will be better governance and more transparency in the exchange.

**Volatility Estimation.** The volatility of the return of the Phisix was determined using the GARCH (1,1). The GARCH (1,1) specification was used since it has been shown to be a parsimonious representation of conditional variance that adequately fits many economic time series (e.g., Bollerslev, 1987). A succinct measure of the persistence of variance as measured by GARCH is the  $\Sigma(\alpha_i + \beta_i) < 1$  (where  $i > 0$ ): as this sum approaches unity, the

greater is the persistence of stocks to volatility. (Lamoureux & Lastrapes, 1990) The GARCH (1,1) results show that there is a large GARCH lag coefficient with Garch of 0.599971 before demutualization and Garch of 0.778186 after demutualization. Table 3 indicates that shocks to conditional variance take a long time to die out so volatility is persistent. This means that the market takes time to absorb the impact of information arrival before adjusting to the true price of the stock. Based on Table 3, the GARCH coefficient is higher after demutualization. As was mentioned earlier, in the GARCH models, volatility clustering is evident. This means that "large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes...." (Kon & Kim, 1994, Fama, 1965 and Mandelbrot, 1963).

On the other hand, large GARCH error coefficients mean that volatility is quick to react to market movements and volatility tend to be spikier. The size of the GARCH lag coefficient and GARCH error coefficient determine the shape of the resulting volatility time series in this case, the returns of the stocks. Based on Table 3, the GARCH error is slightly higher thirty days after demutualization.

**Table 3. GARCH (1,1 Results)**

|                                                            | ARCH (1) | GARCH (1) |
|------------------------------------------------------------|----------|-----------|
| 30 days before demutualization                             | 0.149986 | 0.599971  |
| 30 days after demutualization, 2 days lag after event date | 0.150282 | 0.778186  |

Based on the (G) ARCH models, a volatility forecast is constructed for PHISIX. Appendix 1 presents the forecast using the in-sample for 61 days. The model is evaluated through the use of two forecast error statistics, namely, root mean, squared error, and mean absolute error. The smaller the error, the better the forecasting ability of that model according to that criterion (Eviews User Guide). The results show that the errors are small.

One limitation of the ARCH and GARCH models is that it cannot capture some important features of the data. The most interesting feature is the leverage of asymmetric effect discovered by Black (1976). The exponential ARCH (EGARCH) model of Nelson (1991) was largely influenced by Black's (1976) empirical observation that stock volatility tends to rise following negative returns and to drop following positive returns (Nelson, 1991).

The EGARCH (1,1) of the Phisix shows that that the leverage effect term ( $\gamma$ ) denoted as RES/SQR[GARCH] (1) in the output is negative (Table 4) and statistically different from zero, indicating the existence of the leverage effect to stock return volatility before demutualization. The leverage effect occurs when an unexpected drop in price (bad news) increases predictable volatility more than an unexpected increase in price (good news) of similar magnitude. This effect suggests that a symmetry constraint on the conditional variance function in past  $\varepsilon$ 's is inappropriate (Engle and Ng, 1993). However, after demutualization, there is a positive EGARCH specification for the Phisix which means that there is no leverage effect.

**Table 4. EGARCH (1,1)**

|                                                            | RES/SQR[GARCH](1) | RES/SQR[GARCH](1) | EGARCH (1) |
|------------------------------------------------------------|-------------------|-------------------|------------|
| 30 days before demutualization                             | 0.385298          | 0.421875          | -0.169254  |
| 30 days after demutualization, 2 days lag after event date | -0.649661         | -0.771840         | 0.347969   |

The G(ARCH) model is more advantageous than the standard deviation since it models time-varying volatility as compared to the unconditional standard deviation.

An investor usually does a mean-variance analysis in developing his portfolio. Since reversals do not occur very quickly, it is hard to attribute the changes in volatility to a trading-related phenomenon. Extreme return days occur on average following substantial losses, and jumps in stock prices are intertemporally clustered.

## CONCLUSION

To remain globally competitive, the Philippine Stock Exchange, Inc.(PSE) has to find ways of improving the stock market and have more investor participation. The first step they undertook is to become demutualized. The conversion was made possible through a technical assistance grant from the Australian Agency for International Development, given through the Philippine Australia Governance Facility. As such, the ASX model was used as framework for the demutualization of the PSE.

The study found out that stock return volatility of PHISIX exhibited the same characteristics as what previous studies have found out: (1) volatility is time-varying; (2) returns are leptokurtic, (meaning having fat tails relative to the normal distribution, and tall peaks); (3) skewness; and (4) volatility clustering. With volatility clustering, large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes. This means that it takes at least a lag of one day before the market can fully capture the effects of information arrival and adjust stock prices to its true price. This can be observed in the F-test results and the GARCH (1,1) model of the Phisix's return volatility. It can be noted that there is more volatility after demutualization. Also, as evidenced by the EGARCH (1,1) models, stock return volatility exhibited leverage effect before demutualization.

There are two possible explanations for such observations. First, the Philippine stock exchange has a short trading period, from 9:30 am to 12:00 noon with a ten-minute extension. Although, information arriving during the run-off period can affect volume, volatility is no longer affected since the closing price at 12 noon remains. Secondly, stock return volatility may lag for one or more days because of the price freeze rule. Even though, there is a price freeze rule, production of information continues. Once, the price freeze is lifted, the return volatility for the stock may continue for several days until it reaches its true price.

Although, the GARCH models capture the time-varying volatility, it seemed that investors analysis are limited to the mean-variance. The forecast made was for a short-term, specifically sixty-one days. The short period was necessary to capture investors immediate reaction to information arrival, in this case the demutualization of the PSE.

In conclusion, demutualization of the stock exchange affected the Phisix return volatility. But as mentioned earlier, there are other events which would have affected the volatility of the Phisix, such as the September 11 World Trade Center disaster.

## RECOMMENDATIONS

As mentioned earlier, investors focus mainly on the mean-variance (as represented by the constant standard deviation) analysis, which represents returns (mean) and risk (variance). This is a simple way of determining which stocks to include in an efficient portfolio; however, the GARCH models provide time-varying standard deviation. With the analysis using the time-

varying standard deviation, an investor can combine stocks that may result with minimum risk and expected returns. Also, investors may use the fundamental analysis concentrating mainly on price-earning ratio, and technical analysis, charting price patterns to determine the next support or resistance level. If the investor can take advantage of the short-term forecast, then he/she would have an efficient portfolio. Perhaps, it is about time that investors do their homework and not rely mainly on researches of stock brokerages.

With a demutualized exchange, there are reforms needed to be implemented. First, go through the second phase of demutualization which is the initial public offering of the exchange. Also, there is a need to draft a policy direction that will separate the management of the board of directors and the delineation of their function as suggested by the Securities and Exchange Commission (SEC) to promote better corporate governance.

Based on the talk given by SEC Commissioner Ma. Juanita E. Cueto on "International Best Practice on Corporate Governance" last December 6, 2001 at SEC, "corporate governance was defined as involving a set of relationships between a company's management, its board, its shareholders, and other stakeholders, and provides the structure through which the objectives of the company are set and the means of attaining those and monitoring performance are determined." She gave the characteristics of Equity/Bond Market System (EMS):

"Ownership should be widely dispersed, that is, having many institutional investors. There should be a separation of ownership and management control to avoid agency problems. And an annual general shareholder's meeting be held for continued monitoring and ultimate shareholders' control.

There should be a lot of publicly listed companies and must be willing to establish accounting and disclosure practices. There should also be a established basis for liability of independent external auditors. For the part of the stock market, it should be large and liquid. The securities trading rules should be established to encourage more active investor protection. Also, there is a need to strengthen the self-regulatory powers of the bourse and bring in greater transparency in the market. Of course, the Philippine stock market has still to brace itself on the uncertainties brought about by political and regional factors."

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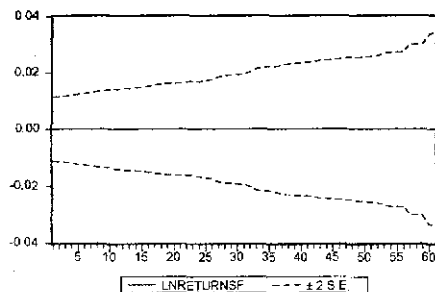
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## APPENDIX A

### GARCH FORECAST

APPENDIX A  
GARCH 1,1  
IN-SAMPLE FORECAST



Forecast: LNRETURNSF  
Actual: LNRETURNS  
Forecast sample: 1 61  
Included observations: 61

|                              |          |
|------------------------------|----------|
| Root Mean Squared Error      | 0.012197 |
| Mean Absolute Error          | 0.008255 |
| Mean Abs. Percent Error      | 100.0000 |
| Theil Inequality Coefficient | 1.000000 |
| Bias Proportion              | 0.059559 |
| Variance Proportion          | 0.940441 |
| Covariance Proportion        | 0.000000 |

